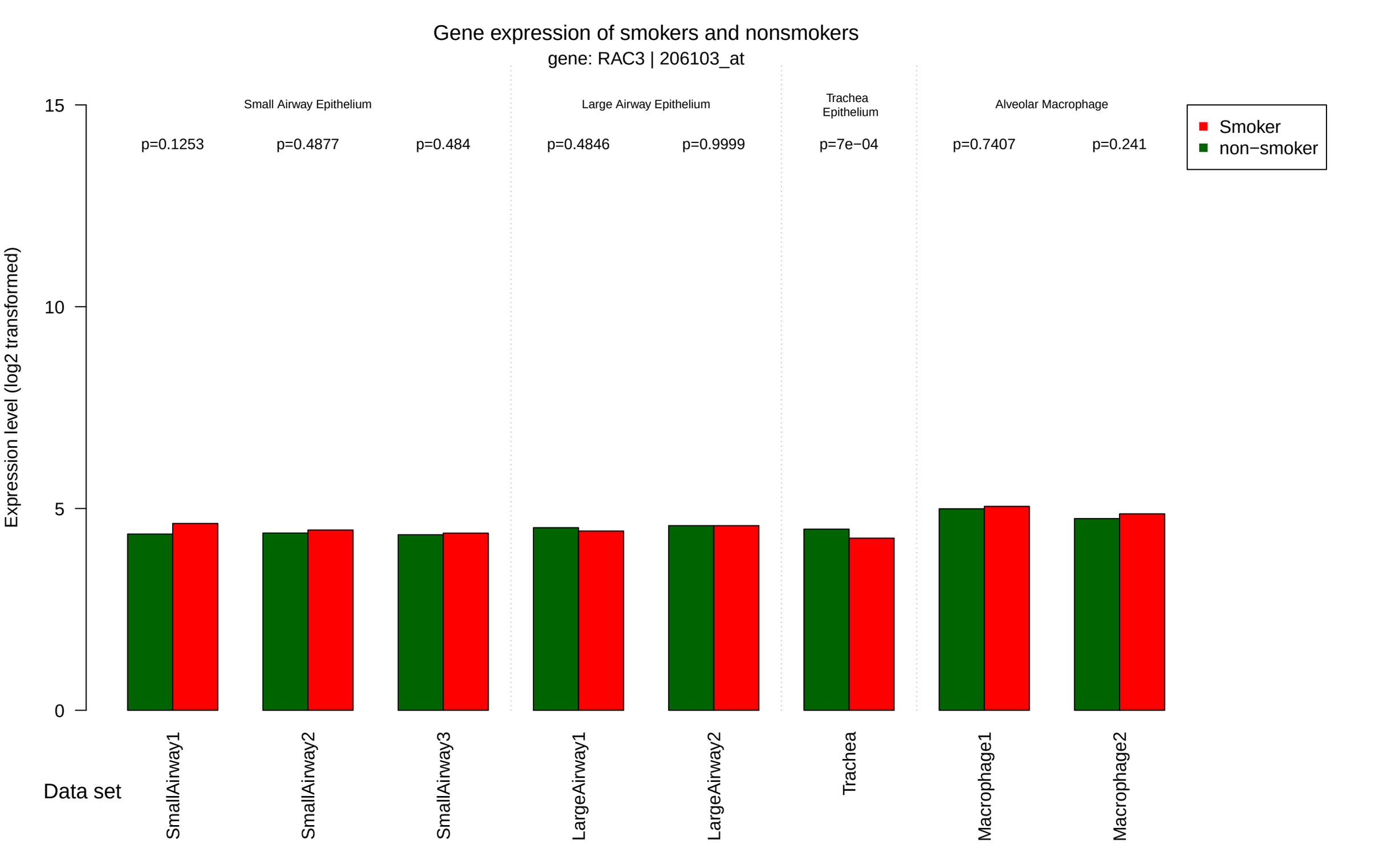
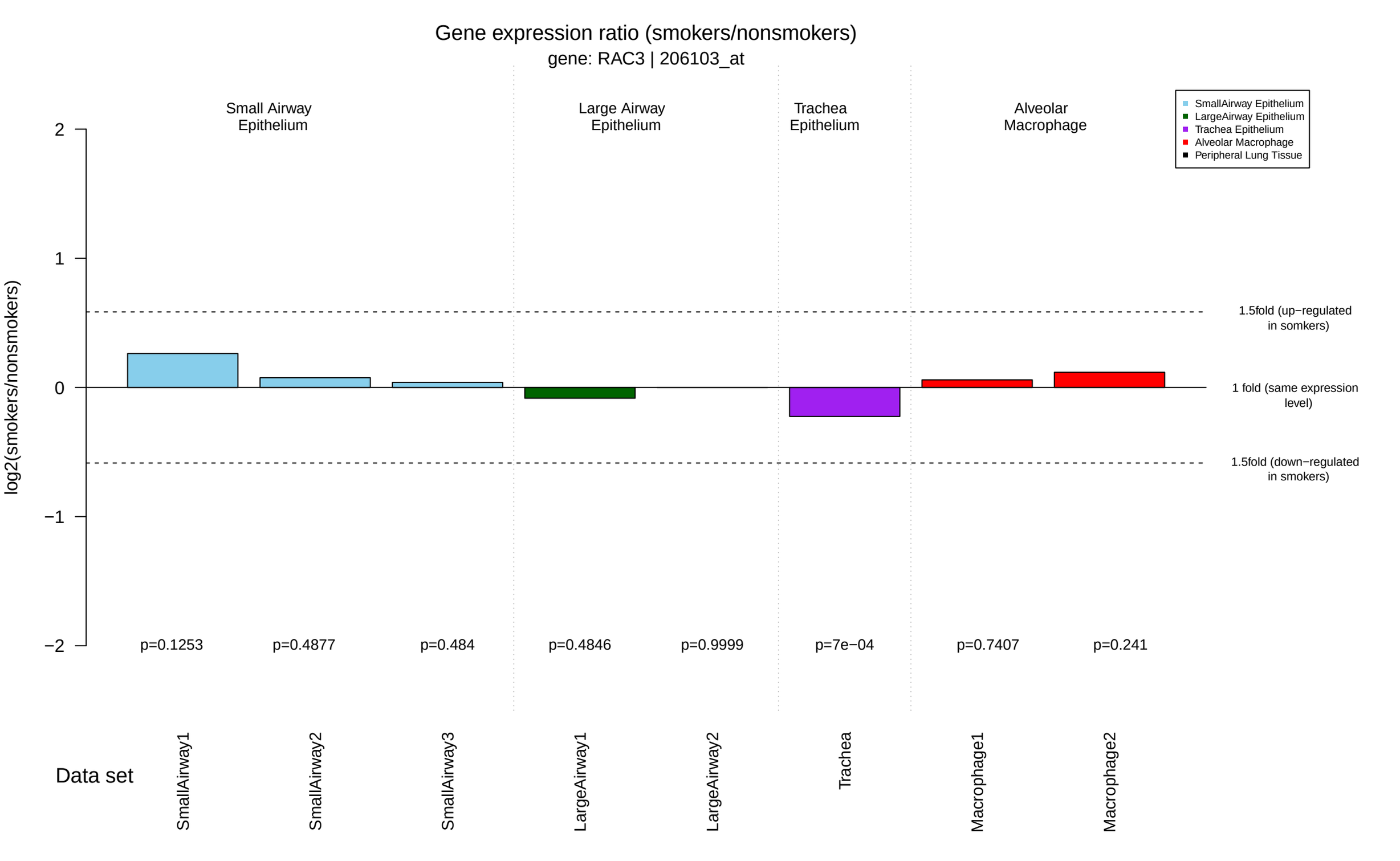
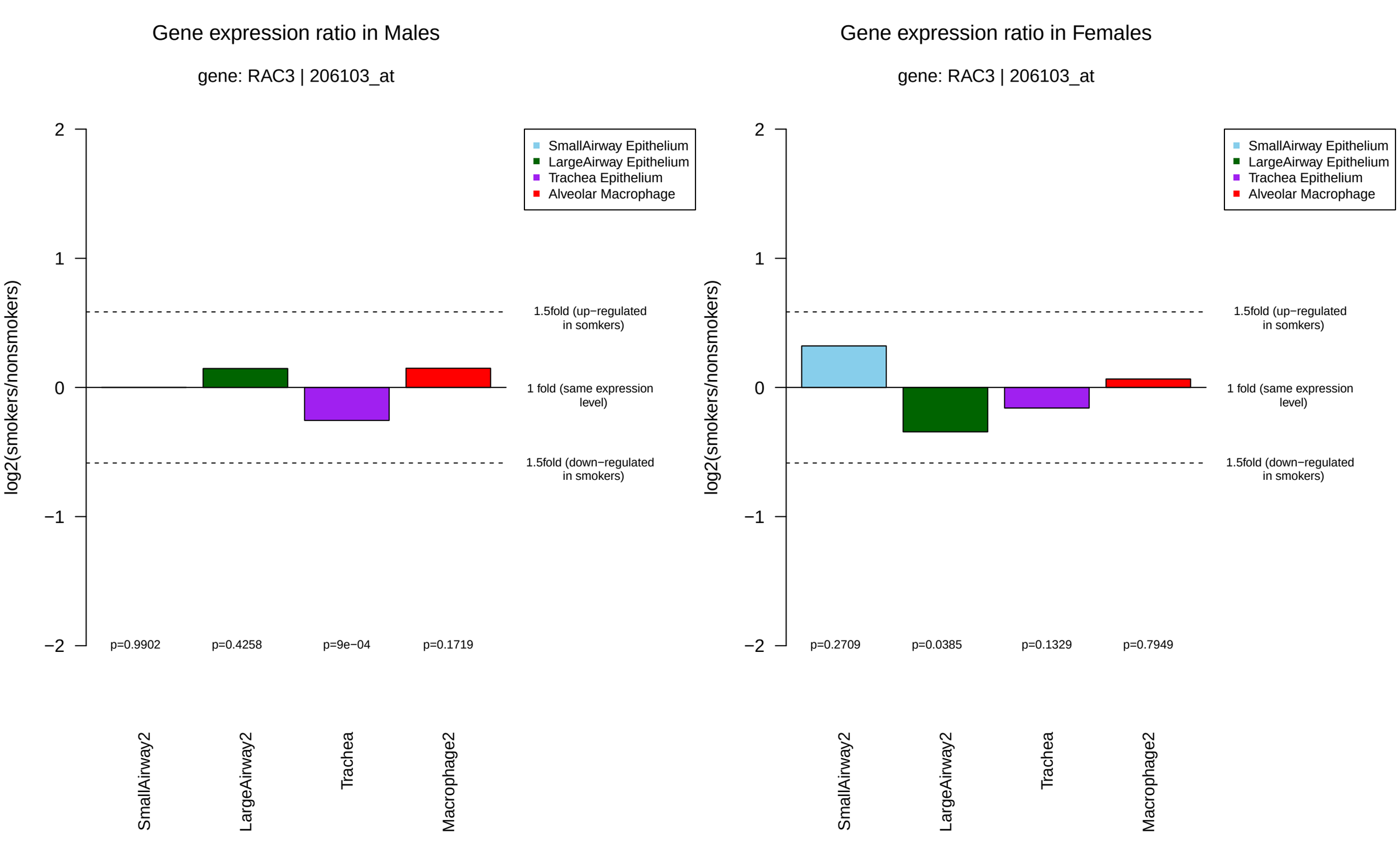
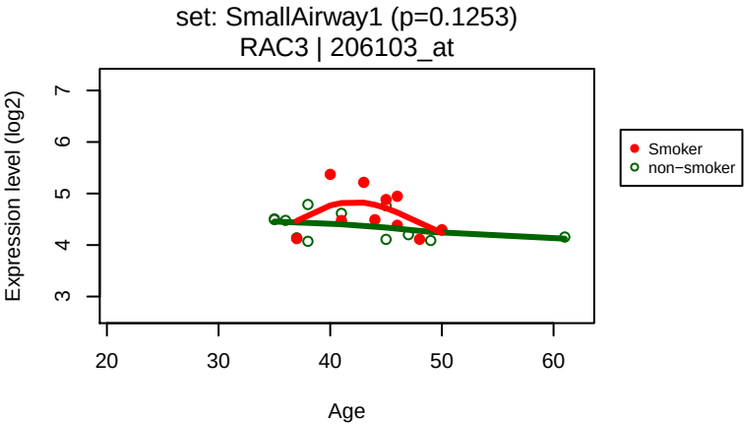
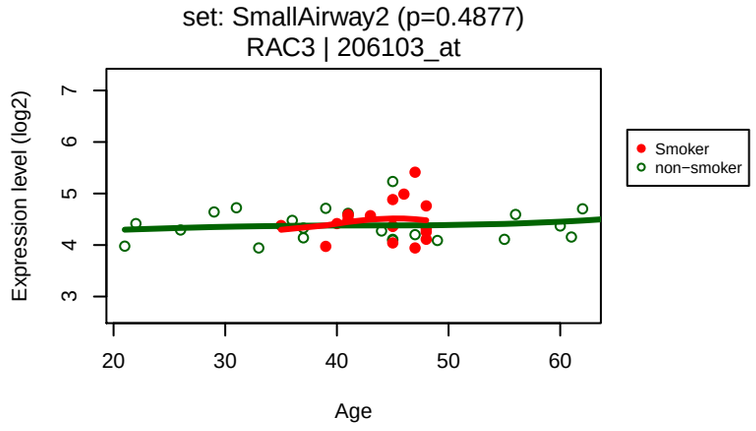
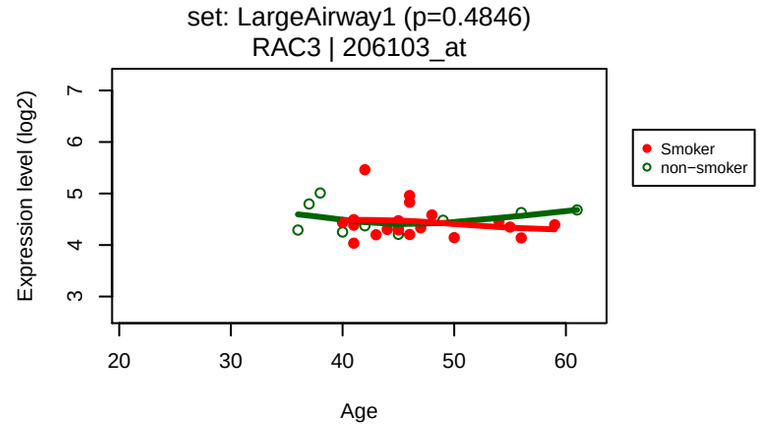

Gene expression of smokers and nonsmokers
gene: RAC3 | 206103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1253
p=0.4877
p=0.484
p=0.4846
p=0.9999
p=7e−04
p=0.7407
p=0.241
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

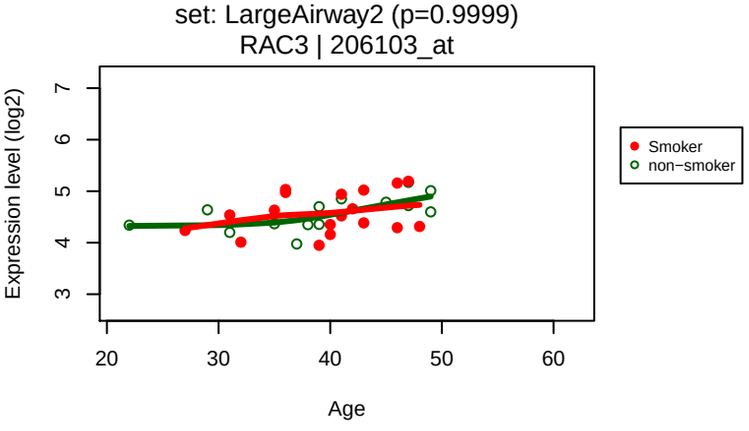

Gene expression ratio (smokers/nonsmokers)
gene: RAC3 | 206103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1253
p=0.4877
p=0.484
p=0.4846
p=0.9999
p=7e−04
p=0.7407
p=0.241
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

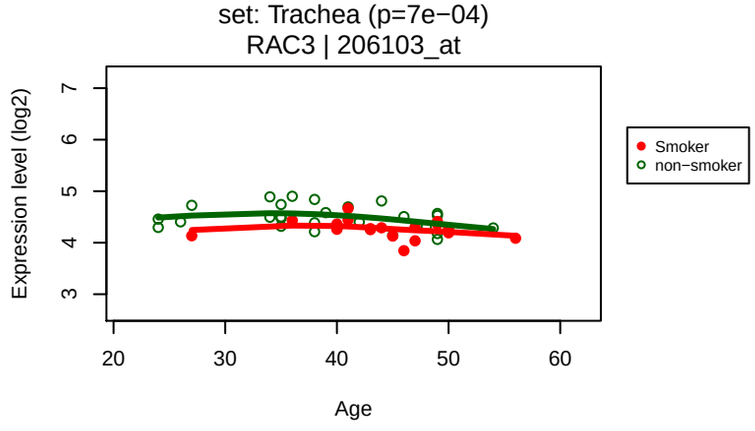

Gene expression ratio in Males
gene: RAC3 | 206103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9902
p=0.4258
p=9e−04
p=0.1719
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAC3 | 206103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2709
p=0.0385
p=0.1329
p=0.7949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

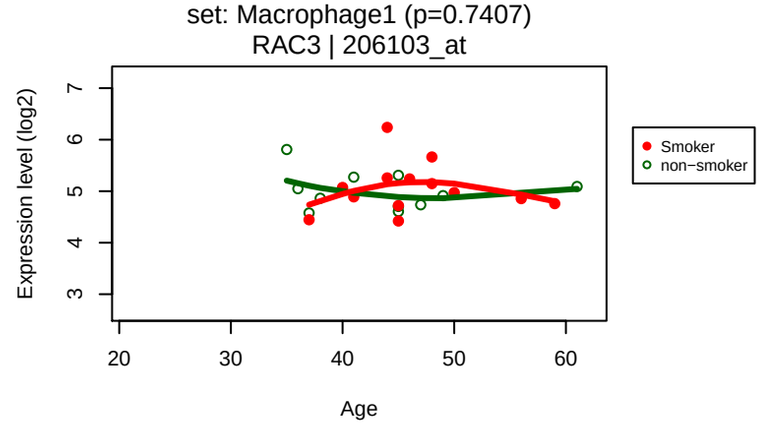

set: SmallAirway1 (p=0.1253)
RAC3 | 206103_at
Expression level (log2)
Age
Smoker
non−smoker

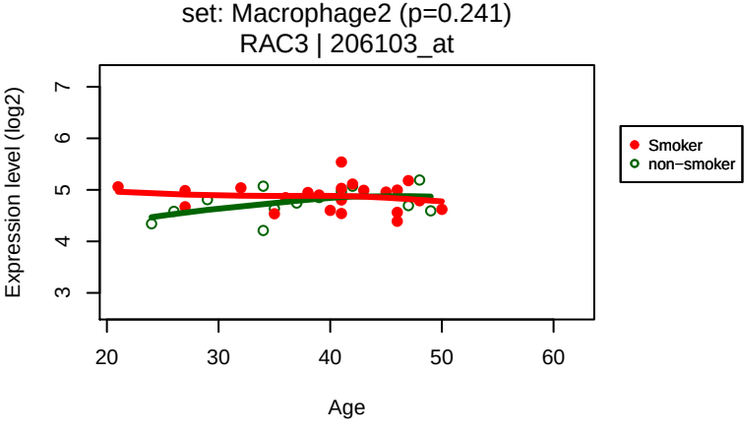

set: SmallAirway2 (p=0.4877)
RAC3 | 206103_at
Expression level (log2)
Age
Smoker
non−smoker

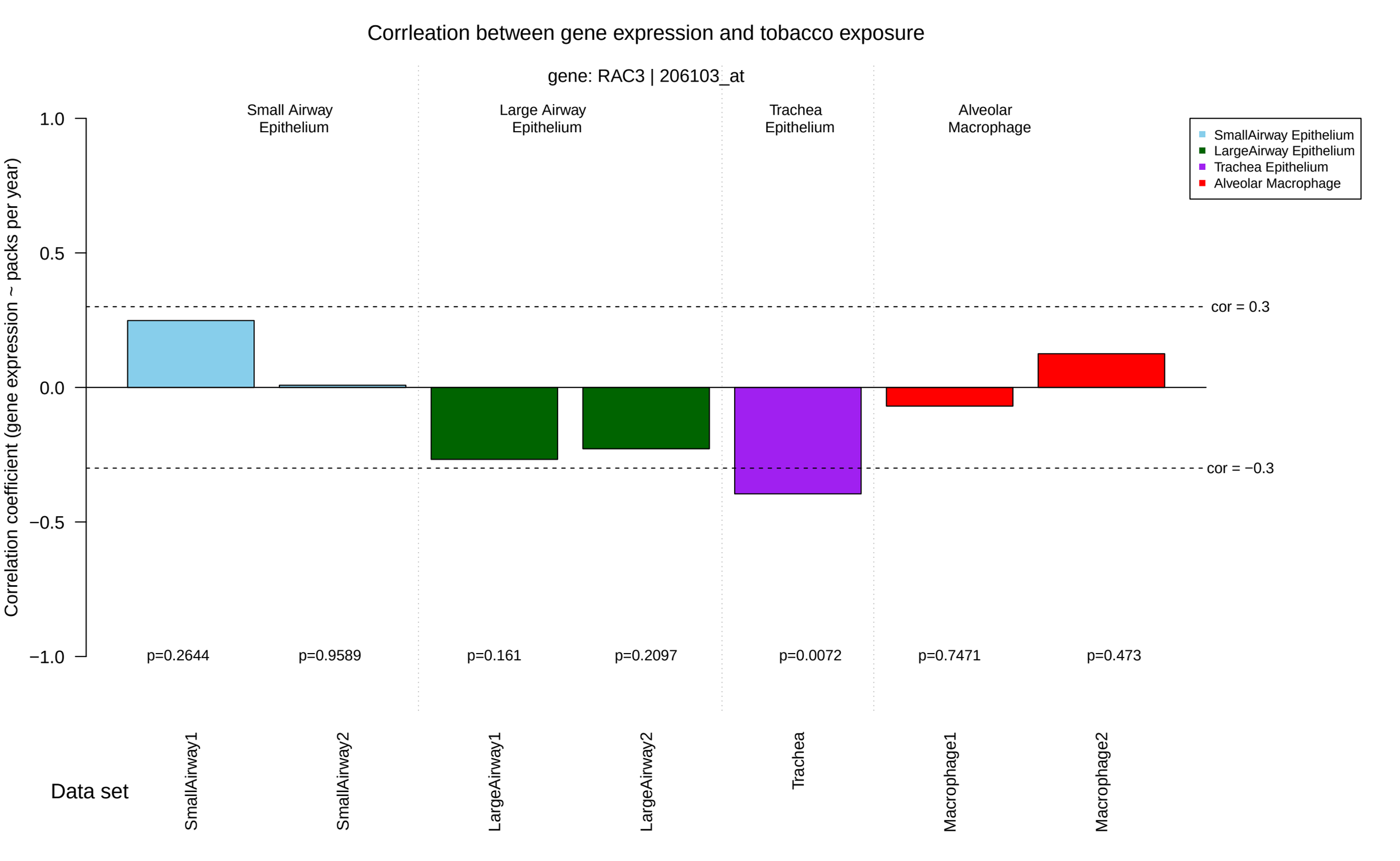

set: LargeAirway1 (p=0.4846)
RAC3 | 206103_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9999)
RAC3 | 206103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
RAC3 | 206103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7407)
RAC3 | 206103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.241)
RAC3 | 206103_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAC3 | 206103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2644
p=0.9589
p=0.161
p=0.2097
p=0.0072
p=0.7471
p=0.473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2